

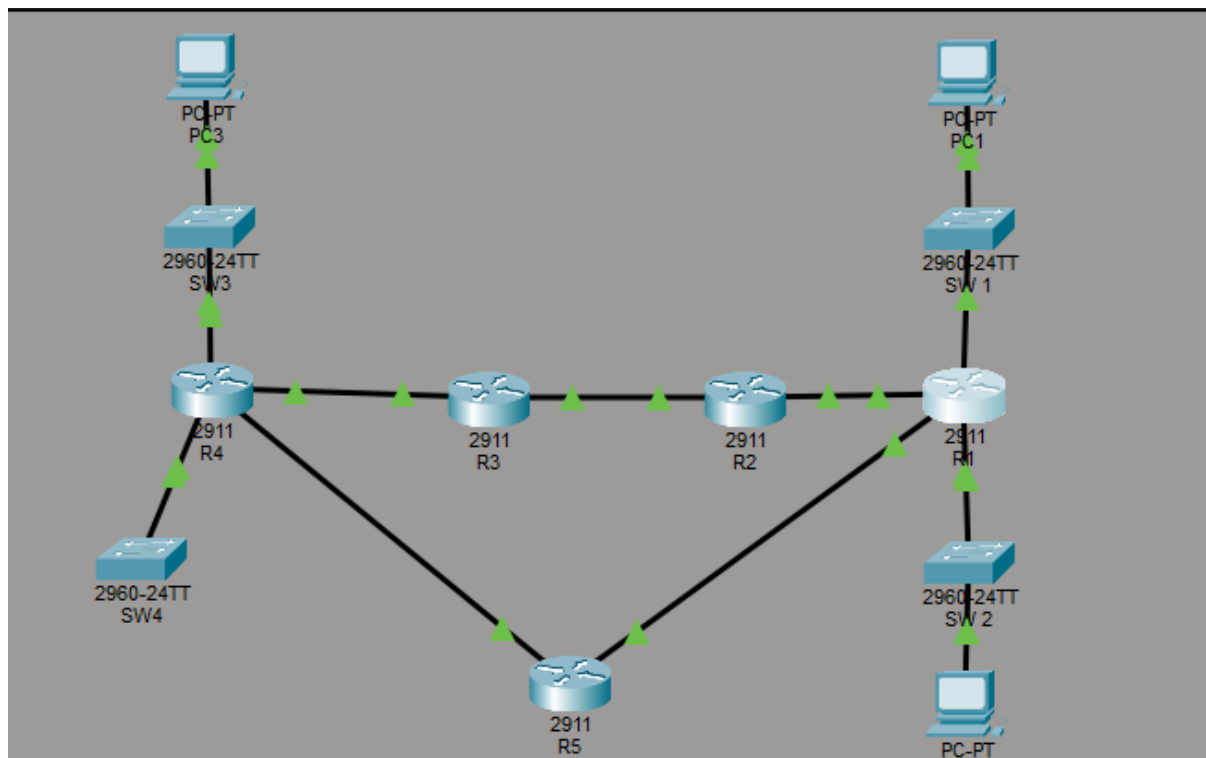


NET512 PRAC WEEK 7

Lerato Modise



Screenshot of the topology



Verify connected

```
Router(config-if)#hostname R1
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 10.0.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#interface GigabitEthernet0/1
R1(config-if)#ip address 10.1.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.1.0/24 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
C       10.1.0.0/24 is directly connected, GigabitEthernet0/1
L       10.1.0.1/32 is directly connected, GigabitEthernet0/1
```

Ping and tracert from PC1 to PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  0  *         *         *         Request timed out.
  1  *         *         *         Request timed out.
  2  *         *         *         Request timed out.
  3  *         *         *         Request timed out.
  4  *         *         *         Request timed out.
  5  *         *         *         Request timed out.
  6  *         *         *         Request timed out.
```

```
R1>enable
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.0.2.0 255.255.255.0 10.0.1.2
R1(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.2
R1(config)#
R1(config)#no ip route 10.1.2.0 255.255.255.0 10.1.0.2
R1(config)#ip route 10.0.0.0 255.255.0.0 10.0.1.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S       10.0.0.0/16 [1/0] via 10.0.1.2
C       10.0.1.0/24 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
S       10.0.2.0/24 [1/0] via 10.0.1.2
C       10.1.0.0/24 is directly connected, GigabitEthernet0/1
L       10.1.0.1/32 is directly connected, GigabitEthernet0/1
```

```
Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 203.0.113.2

Tracing route to 203.0.113.2 over a maximum of 30 hops:

  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
  4  *          *          *          Request timed out.
  5  *          |
```

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S       10.0.0.0/16 [1/0] via 10.0.1.2
C       10.0.1.0/24 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
S       10.0.2.0/24 [1/0] via 10.0.1.2
C       10.1.0.0/24 is directly connected, GigabitEthernet0/1
L       10.1.0.1/32 is directly connected, GigabitEthernet0/1

```

```

R1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int fa0/0
%Invalid interface type and number
R1(config)#GigabitEthernet0/0
^
% Invalid input detected at '^' marker.

R1(config)#ip route 203.0.113.0 255.255.255.0 10.0.3.2
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.1.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.1.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
R1(config)#

```

Reflection Questions

Do you see routes for every network R1 is directly connected to?

Yes, because connected routes are automatically added when interfaces are configured.

Why does PC1 lose reachability to PC3 when static routes are removed?

Without static routes R1 has no knowledge of remote subnets so packets cannot be forwarded.

Why can a single summary route replace multiple /24 static routes?